

JP Seismic Tomography - Introduction

A plain-English description of the experiment and its purpose

1. What this folder contains

This folder contains the material used to reproduce the Japan seismic-tomography benchmark from the dissertation. The numerical program is `JP_Seismic_Tomography.py`, and the plotting program is `plot_JP.py`. The purpose of the two programs is deliberately kept separate: the first performs the numerical experiment and records the measurements, while the second reads those measurements and produces the figures.

The experiment uses a real sparse matrix from the SuiteSparse Matrix Collection. It is not a synthetic matrix generated by the benchmark code. The matrix is associated with the Harvard Seismology group and is identified in the code as `Harvard_Seismology/JP`.

2. What the JP matrix represents

The JP matrix is a forward operator from a linearised seismic-tomography problem. In practical terms, seismic tomography tries to infer information about the Earth's interior from measurements of seismic waves. A forward model describes how a proposed set of subsurface parameters would affect the measurements. The matrix A represents that forward relationship in linearised form.

The matrix is very large and sparse. It has 87,616 rows, 67,320 columns, and 13,734,559 non-zero entries. The dimensions can be written as $A \in \mathbb{R}^{(87616 \times 67320)}$. The large number of zero entries is important: the experiment is intended to work with the sparse structure rather than turning the matrix into a dense 87,616 by 67,320 array.

The rows correspond to equations or measurements in the linearised model, while the columns correspond to the unknown model parameters. There are therefore more equations than unknowns. The benchmark treats the problem as a large rectangular least-squares problem.

3. The mathematical problem used here

The numerical experiment solves the regularised least-squares problem

$$\text{minimise over } x: \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

The first term measures how closely the computed vector x satisfies the data equations. The second term is the ridge-regularisation penalty. The parameter λ controls the strength of that penalty. Small λ values put more emphasis on fitting the equations; larger values place more weight on the regularisation term.

The original SuiteSparse release used here does not provide the observed seismic right-hand side required to reproduce the original geological inversion. For that reason, the benchmark constructs a controlled right-hand side using $b = A @ \text{ones}$. This means the experiment is a solver benchmark performed on a real scientific matrix; it should not be described as a reconstruction from the original observed seismic data.

4. The four solvers

Four iterative approaches are compared. They are all given the same matrix A , the same constructed vector b , and the same regularisation value for a particular run.

Solver	How it is used
CG	Conjugate Gradient applied to the regularised normal-equation operator. The operator is evaluated matrix-free; A-transpose-A is not stored as a separate matrix.
PCG	Conjugate Gradient with a Jacobi diagonal preconditioner. The normal-equation operator remains matrix-free.
LSQR	Damped least-squares iteration using the sparse matrix A. The code does not explicitly construct A-transpose-A.
LSMR	Damped least-squares iteration using the sparse matrix A. The code does not explicitly construct A-transpose-A.

5. Why the benchmark compares these methods

The main question is not simply which solver is fastest. The benchmark records runtime, iteration count, residual measures, objective value and convergence status so that speed can be considered alongside numerical behaviour. A solver that finishes quickly but leaves a substantially different residual is not equivalent to one that reaches a comparable numerical state more slowly.

The comparison is repeated over a range of regularisation values. This is important because regularisation changes the numerical problem itself. The dissertation results show that solver rankings can change as lambda changes; there is therefore no single lambda at which the entire comparison should be judged.

6. Regularisation sweep in the archived benchmark

The archived dissertation run used the following explicit lambda values:

```
1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1, 10
```

The current repository version of JP_Seismic_Tomography.py also supports an adaptive lambda mode based on multiples of the mean diagonal of A-transpose-A. Its defaults are 1e-3, 1e-2, 1e-1, 1, 10, 100 and 1000. For exact reproduction of the dissertation's reported JP sweep, the explicit lambda option should be used rather than silently relying on those adaptive defaults. The Run Instructions document gives the command.

7. What is measured

For every solver, lambda value and repeated run, the numerical program records the elapsed runtime, number of iterations, residual quantities, objective value and convergence status. It also records the matrix dimensions, non-zero count, density, regularisation information, tolerance and iteration limit. These details are written to CSV files so that the plots can be regenerated without rerunning the expensive benchmark.

8. What the plotting stage does

plot_JP.py does not solve the tomography problem. It reads the CSV files produced by the numerical program and creates the figures used to inspect the benchmark. The supplied plotting code produces ten PNG figures, covering runtime, iterations, relative residual, relative normal-equation residual, objective value, average solver performance, LSMR speedup and convergence success.

9. Interpreting the experiment

The JP experiment should be read as a controlled comparison of numerical methods on a real, large sparse scientific matrix. The use of $b = A @ \text{ones}$ makes the right-hand side reproducible and

removes dependence on unavailable observational data. It does not turn the experiment into a physical reconstruction of Japan.

The most useful outcome is therefore the relationship between solver, regularisation, runtime, iteration count and residual behaviour. In the dissertation, stronger regularisation reduced computational effort at the upper end of the sweep, while also changing the residual behaviour. This is precisely why the full sweep is retained rather than selecting a single convenient value.

10. Repository workflow

- Keep JP_Seismic_Tomography.py and plot_JP.py together.
- Run the numerical benchmark first and retain the generated CSV files until the plots have been produced and checked.
- Run plot_JP.py only after the required CSV files exist.
- The generated CSV and PNG files are outputs of the experiment and do not need to be committed to the repository if the repository is intended to contain reproducible code and documentation only.
- Record the computational environment and resource allocation when reproducing timing results, because wall-clock measurements depend on the machine and allocation used.